

Danish Abbas Khan

Physics Undergraduate

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EDUCATION

Pakistan Institute of Engineering and Applied Sciences (PIEAS)

Bachelors of Science in Physics:

Sep 2022 – July 2026

CGPA: 3.69/4

Class Rank: 3/40

Thesis: Study of Cosmic Ray Atmospheric Ionization

Relevant Courses: Introduction to Particle Physics, QFT, Quantum Mechanics, Computational Physics, Fundamentals of Machine Learning, Nuclear Physics, Mathematical Methods of Physics

Govt. Degree College For Boys Garhmaharaja

Intermediate - Pre-Medical:

2019 – 2022

Board of Intermediate & Secondary Education, Faisalabad

Class Rank: 1/75

Percentage: 97%

RESEARCH EXPERIENCE

Undergraduate Dissertation (BS Physics)

Sep 2025 – July 2026

PIEAS, Islamabad, Pakistan

Thesis: Study of Cosmic Rays Atmospheric Ionization

- Conducted macroscopic Monte Carlo simulations of extensive air showers using **CORSIKA** in a **Linux** environment, integrating real-time **GDAS** atmospheric profiles to track 10,000 primary particles and accurately characterize the longitudinal Pfozter maximum.
- Transitioned the core simulation engine to **C++/Geant4** to engineer a high-precision $10 \times 10 \times 12 \text{ km}^3$ atmospheric volume, heavily optimizing the geometry into 120 discrete 100-meter strata utilizing empirical thermodynamic models (Tetens equation).
- Developed automated data analysis pipelines utilizing **Python** and **R** to parse massive, unformatted binary arrays from both simulation frameworks, rendering complex spatial energy deposition heatmaps and macroscopic cascade profiles.
- Managed intense computational workloads across a remote university server cluster (70+ hours runtime), executing 1,000,000 primary events via the **CRY** library with strict PRNG synchronization and a 1 cm step limit to map atomic-level scattering devoid of stochastic noise.

INTERNSHIPS

Detector Response and Visualization Using the GEANT4 Toolkit

Jun 2025 – Aug 2025

Research Intern | Experimental High Energy Physics Dept., National Centre for Physics (NCP)

- Modified GEANT4 B1 example to simulate particle interactions and extract detector-level data (energy, momentum, position).
- Enhanced visualization using geometry control and track rendering, gaining practical experience in simulation workflows.

Experimental and Computational Study of Chaotic Systems

Jul 2024 – Sep 2024

Research Intern (Nonlinear Motion) | PIEAS

- Modeled chaotic motion of a magnetic pendulum by solving ODEs in Python and analyzing sensitivity to initial conditions.
- Integrated Arduino-based data acquisition with basic Mathematica simulations for validation and visualization.

HONORS AND AWARDS

- Awarded a laptop under the Prime Minister's Youth Laptop Scheme on the basis of academic merit, Government of Pakistan (Nov, 2024)
- Recipient of benevolent fund academic support in recognition of strong academic performance, provision of financial assistance toward ongoing studies (Nov, 2025)
- Second-place award for team presentation at the 14th School on LHC Physics, National Centre for Physics (NCP), Islamabad, Pakistan (Aug, 2025)
Competition on LHC data analysis topics; team recognized for outstanding presentation skills.

PROJECTS

Brain Tumor Classification Using Two-Stream Hybrid Quantum CNNs

March 2026 – May 2026

Academic Research Project | PIEAS

- Architected a Two-Stream HQ-CNN fusing an unfrozen EfficientNetB0 classical backbone with a 4-qubit parameterized quantum circuit (PennyLane) for advanced non-linear edge detection.
- Engineered a robust preprocessing pipeline using OpenCV for smart contour-cropping of the publicly benchmarked Figshare brain MRI dataset, applying data-safe stratified augmentation to balance it to 7,500 samples.
- Developed a fast vectorized quantum convolution algorithm to eliminate training bottlenecks, ultimately achieving a highly robust 94.05% accuracy across 5-fold cross-validation.

Unemployment vs. Crime Rate Data Analysis in the USA (2000–2014)

Apr 2024 – May 2024

Collaborative Data Analysis Project | PIEAS

- Analyzed 15-year dataset to evaluate correlation between unemployment and crime rates across U.S. states.
- Performed statistical analysis and visualized trends using Excel, $\text{\LaTeX}$, and presentation tools.

Sales Analysis and Dashboard Creation using Microsoft Excel

Feb 2023 – Mar 2023

Independent Project | Coursera Certification

- Developed interactive dashboards using pivot tables and charts to analyze regional and monthly sales performance.
- Applied data cleaning and formatting techniques; achieved course certification with honors distinction.

SCHOOLS AND CONFERENCES

- 14th School on LHC Physics, National Centre for Physics (NCP), Islamabad, Pakistan (Aug 2025)
High-energy physics training on LHC data analysis, accelerators, detectors, and statistical techniques.
- Advanced Training in High-Energy Physics (ATHEP), National Centre for Physics (NCP), Islamabad, Pakistan (Feb 2026)
Lectures and hands-on sessions on particle physics, cosmic rays, and data analysis techniques.
- International School on Physics & Allied Disciplines (ISPAD-2026), NCP, Islamabad, Pakistan (Apr 2026)
Specialized focus on Atmospheric Physics, weather modification strategies, and numerical modeling for cloud seeding.

TECHNICAL SKILLS

- **Computational Engines:** CORSIKA, GEANT4, CRY
- **Languages & Data Analysis Frameworks:** Python, C, R, ROOT (PyROOT), Wolfram Mathematica
- **Libraries & Tools:** TensorFlow, PyTorch, PennyLane (Quantum ML), OpenCV, SimpleITK, Git, $\text{\LaTeX}$, Linux (Ubuntu)
- **Professional & Interpersonal:** Scientific Writing, Statistical Modeling, Team Leadership, Problem Solving

EXTRACURRICULAR ACTIVITIES

- Supported event management teams with coordination and logistics.
- Core task force member of Physics Society for PIEAS (PSP).
- Helped organize and manage discussions and events such as *Beyond Boundaries*, promoting open dialogue on diverse academic and social topics.